

EC5011-DIGITAL SIGNAL
PROCESSING
AVIAN ACOUSTICS -ASSIGNMENT 1

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GROUP EG13

SEMESTER 5

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HARNESSING BIRDSONG FOR ENVIRONMENTAL MONITORING

INTRODUCTION:

One of the most important markers of ecosystem health and biodiversity is bird song. In order to provide a thorough environmental monitoring system, this idea seeks to analyse bird vocalizations using sophisticated digital signal processing techniques. We can help with conservation efforts and provide real-time insights into environmental changes by analyzing variations in bird songs to identify ecological alterations.



CONCEPT:

Bird singing signals contain complex acoustic information that varies by species, behavior, season, and environmental condition. By analyzing these signals using digital signal processing techniques, we can build a **real-time environmental intelligence system** that passively listens to nature and gives us meaningful insights.

REAL WORLD APPLICATION:

- ✓ **Biodiversity Assessment:** Monitoring bird populations can provide insights into species richness, habitat quality, and ecosystem health. Vocalizations vary among species, allowing for species identification and abundance estimation.
- ✓ **Habitat Monitoring:** Bird vocalization studies are useful for identifying key nesting locations, determining the suitability of habitat, and identifying habitat problems such as urban encroachment and deforestation.
- ✓ **Ecosystem Dynamics:** Because bird vocalizations reflect alterations in seasonal patterns, habitat degradation, and climate variability, they can be used as early warning indicators of ecosystem disruptions.
- ✓ **Urban Ecology:** Analysis of urban bird communities helps guide planning decisions, evaluates how urbanization affects wildlife, and promotes the creation of green infrastructure to increase biodiversity in urban areas.

ANALYSIS APPROACH:

Analyzing bird singing signals in both the temporal and frequency domains can yield comprehensive environmental knowledge.

Time Domain Analysis: Bird vocalization reveal important information on species behaviour and habitat use through temporal patterns, including daily and seasonal activity cycles, nesting habits, and territorial displays.

Frequency Domain Analysis: Identification of species, habitat distinction, and evaluation of environmental acoustics are made possible by the spectral characteristics of bird songs, such as frequency modulation, pitch variation, and harmonic structure.

PROCEDURE:

How to Record Bird singing Signals;

- **Choosing Recording Equipment:** Make use of a top-notch digital recorder or a smartphone with a dependable microphone that can capture crisp audio in a variety of environmental settings.
- **Picking recording locations,** consider habitat types, bird species variety, and accessibility. Recording is best done in the early morning and late afternoon, when bird activity is usually at its highest.
- **Recorder Positioning:** To improve the collection of bird vocalizations and minimize background noise, position the recorder at the right height and angle. To record consistently and without interruption, use a tripod or solid mount.
- **Configuring Recording Settings:** To get the best audio quality while balancing storage efficiency, change parameters like sample rate, bit depth, and file format (such as WAV).
- **Conducting recording sessions,** make sure there is little wind or rain and that the weather is peaceful. Timer features can be used to automatically start and stop sessions, or they can be started manually.
- **Verification After Recording:** Examine the recorded files to determine the audio quality and spot any distortion or undesired noise. Cut recordings so that only the pertinent parts of the birds chirping are heard.

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Editor - C:\Users\Prabodha\Desktop\2022e116_Task01\EC5011_2022e116_task01.m
EC5011_2022e116_task01.m x +
1 %2022e116
2 %EC5011
3 %task01
4 % Load the .wav file
5 [x, Fs] = audioread('bird_singing.wav');
6
7 % Plot in time domain
8 L = length(x);
9 t = (0:L-1)/Fs*1000;
10 figure;
11 plot(t, x);
12 xlabel('Time (ms)');
13 ylabel('Amplitude');
14 title('Time Domain Signal of Birdsong');
15
16 % Plot in frequency domain
17 Y = abs(fft(x));
18 Ly = length(Y);
19 f = (0:Ly/2-1)*Fs/Ly;
20 figure;
21 plot(f, Y(1:Ly/2));
22 xlabel('Frequency (Hz)');
23 ylabel('Magnitude');
24 title('Frequency Spectrum of Birdsong');
25
26 % Extract MFCCs (Requires Audio Toolbox)
27 % Convert stereo to mono if needed
28 if size(x,2) > 1
29     x = mean(x, 2);
30 end
31
32 % Use mfcc function from Audio Toolbox
33 [coeffs, delta, deltaDelta, loc] = mfcc(x, Fs);
34
35 % Plot MFCCs
36 figure;
37 imagesc(loc, 1:size(coeffs,2), coeffs');
38 axis xy;
39 xlabel('Time (s)');
40 ylabel('MFCC Coefficient Index');
41 title('MFCCs of Birdsong');
42 colorbar;
43
44
```

FIGURE 01:CODE OF BIRD SINGING SIGNAL REPRESENTATION

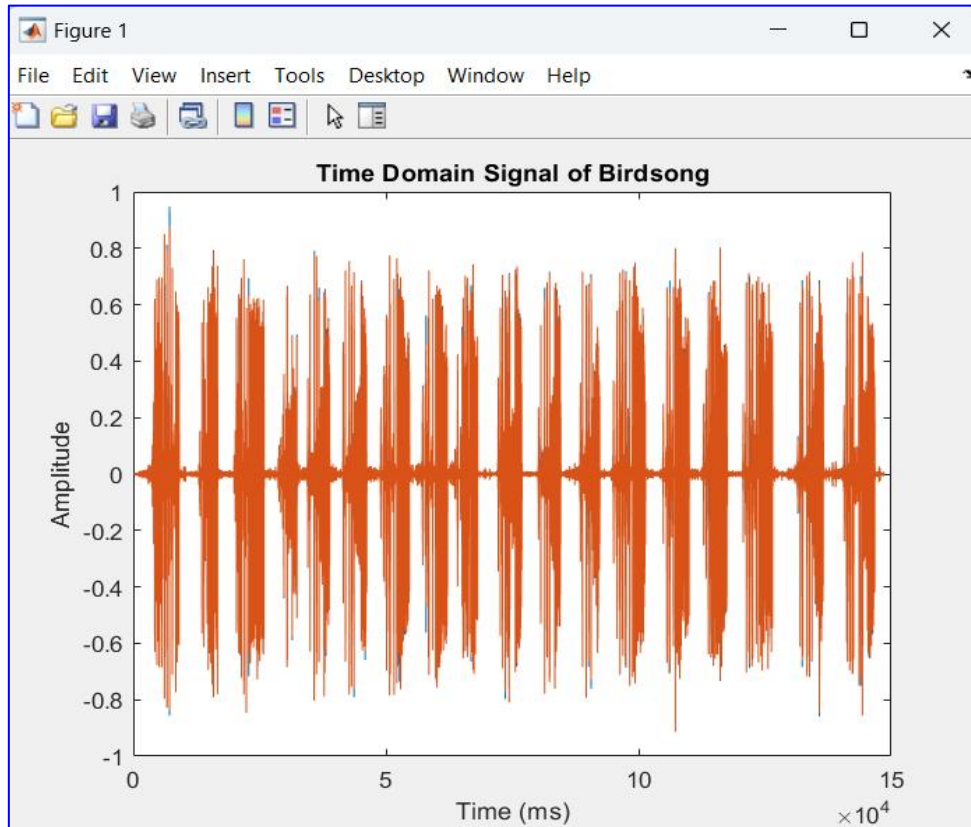


FIGURE 02:TIME DOMAIN SIGNAL OF BIRDSONG

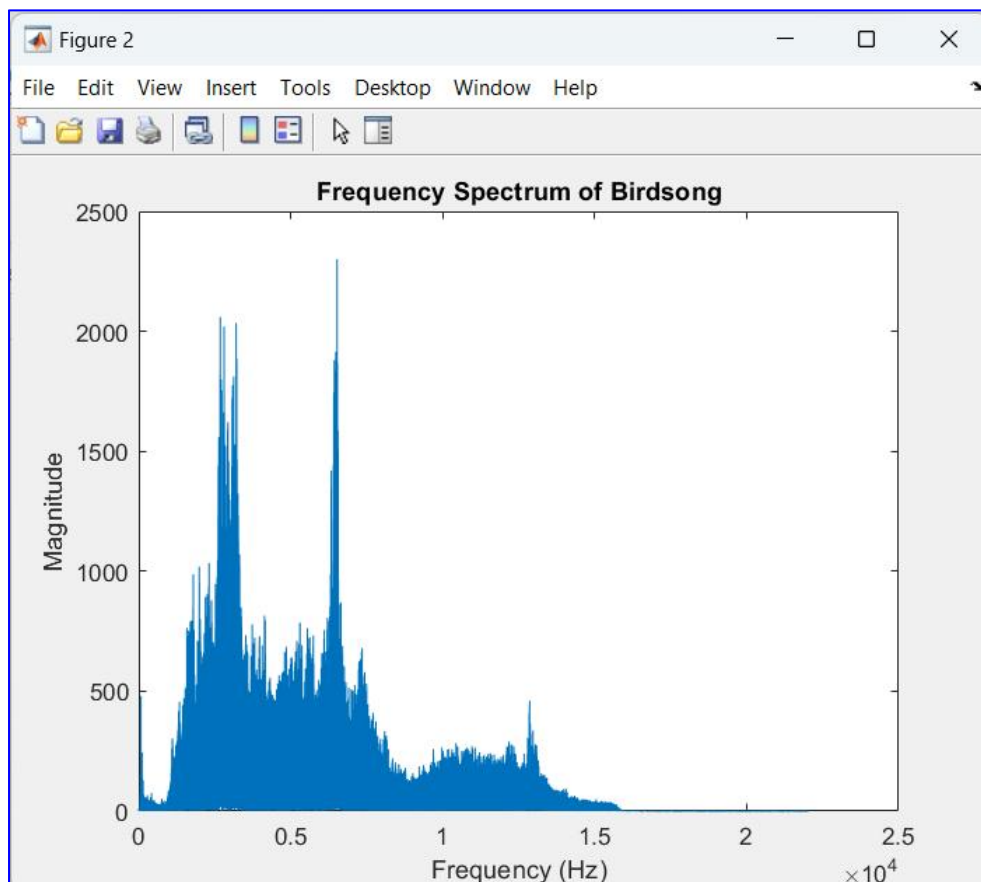


FIGURE 03:FREQUENCY SPECTRUM OF BIRDSONG

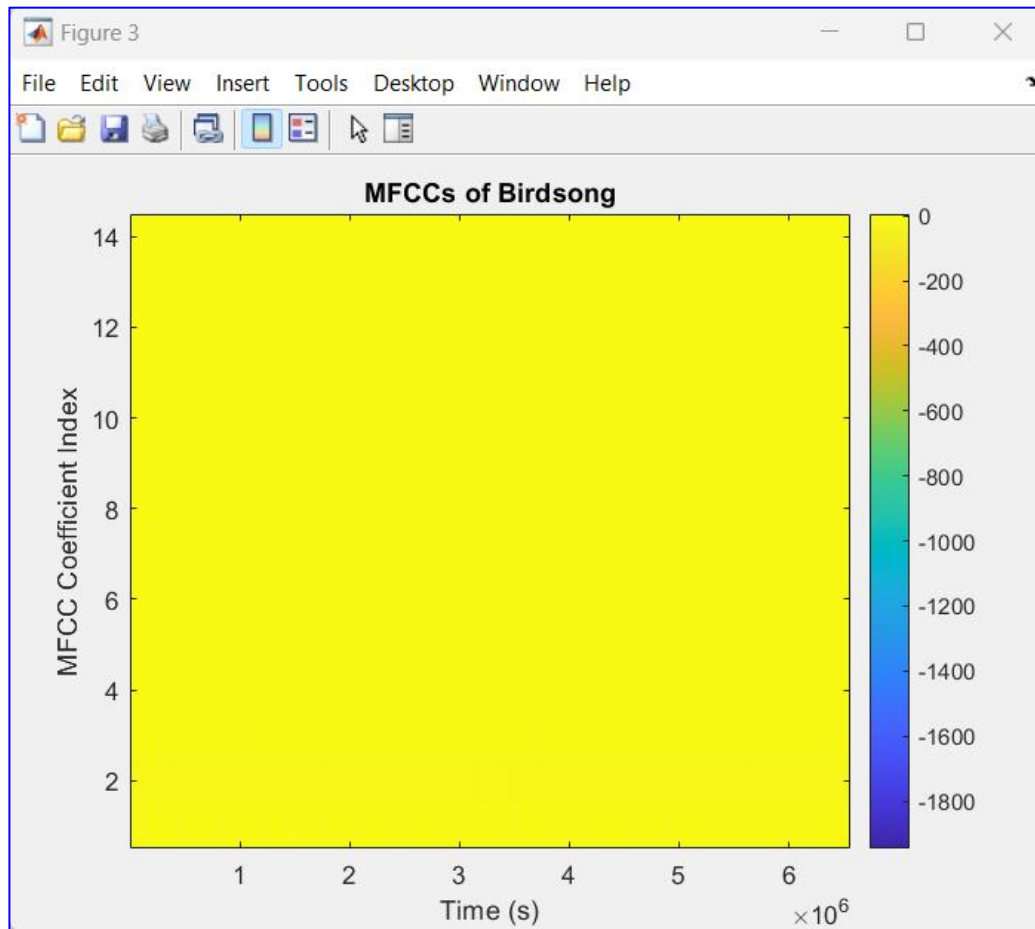


FIGURE 04:MFCCs OF BIRDSONG

PROPOSAL DETAILS

Using Bird Singing Signals to Track the Environment

By combining technological, ecological, and community driven methods, this proposal offers a thorough plan for using bird song signals as a tool for environmental monitoring:

Signal Processing Algorithms: Create sophisticated algorithms that analyse audio recordings, identify species, and extract bird vocalizations. To guarantee precise species identification and signal isolation, methods will involve the examination of spectrograms, frequency distributions, and temporal vocalization patterns.

Ecological Modelling: To forecast species distributions, analyse habitat connectivity, and assess how environmental changes affect avian populations, incorporate data from bird vocalization into ecological models. The prediction ability and ecological relevance of biodiversity evaluations are improved by this integration.

Sensor Networks: Install self-sufficient acoustic monitoring stations with data loggers and microphones in a variety of settings, including urban parks, wetlands, woods, and agricultural regions. Large scale, ongoing data collecting on bird vocalizations will be made possible by these sensor networks.

Community Science Initiatives: Use internet resources, educational programs, and interactive workshops to include citizen scientists and local communities. This cooperative strategy improves data gathering, raises environmental consciousness, and fortifies backing for conservation studies.

Design interactive dashboards and analytical tools to visualize data on bird vocalizations, pinpoint hotspots for biodiversity, assess conservation priorities, and facilitate evidence-based environmental management decision making.

ECONOMIC BENEFITS:

Improvement of Ecosystem Services: Projects to conserve biodiversity and restore habitat support essential ecosystem services like pollination, seed distribution, and natural pest control. Agriculture, forestry, and long-term ecological stability all depend on these services.

Informed Land Management Practices: Important ecological data from acoustic monitoring of bird populations is used to guide land-use decisions in forestry, urban planning, and agriculture, among other industries. This encourages the adoption of sustainable methods that strike a balance between environmental stewardship and development.

Encouragement of Nature-Based Economies and Ecotourism:

Promoting areas with a lot of birds, including wildlife refuges, birding routes, and bird festivals, can increase ecotourism. Through eco-friendly companies, housing, educational initiatives, and guided tours, this generates economic prospects for nearby communities.

POTENTIAL CUSTOMERS:

- Environmental Agencies and Conservation Organizations
- Urban Planners and Landscape Architects
- Agriculture and Forestry Sectors
- Eco-Tourism Operators and Wildlife Reserves
- Academic and Research Institutions
- Wildlife and National Park Authorities
- Government Environmental Departments
- Technology and AI Companies
- NGOs and Community-Based Organizations
- Environmental Consulting Firms
- Climate Change and Resilience Planning Bodies
- Public Health and Urban Wellness Initiatives

CONCLUSION:

A complete way to evaluate ecosystem health, biodiversity, and climate impacts is to incorporate bird song into environmental monitoring. Bird vocalizations are useful bio indicators for conservation efforts because they reflect habitat changes. This proposal offers a methodical way to use sensor networks, data analytic, and sophisticated acoustic technologies to harness these natural signals. It highlights the value of active community involvement and interdisciplinary cooperation between ecologists, engineers, and data scientists. The project promotes inclusivity and environmental awareness by engaging local communities and citizen scientists. By encouraging biodiversity conservation and facilitating prompt interventions, the framework advances sustainable development goals. In the end, using bird songs improves our ability to comprehend, safeguard, and manage natural ecosystems sustainably in a constantly shifting environment.